

CAMBRIDGE INTERNATIONAL AS LEVEL BIOLOGY (9700)

TOPIC 3 – ENZYMES

Mode of Action, Enzyme Kinetics, V_{max} , K_m , Inhibition & Immobilisation

CANDIDATE

Hamna (Grade A* Scholar)

SPECIFICATION

Cambridge 9700 AS Level

TOTAL MARKS / QUESTIONS

220 Marks · 50 Questions

EXAMINATION STRUCTURE & 80/20 QUESTION TARIFF

- Section A – Extended High-Tariff Questions: 20 × 6 Marks (120 Marks total) [Multi-part synoptic evaluation]
- Section B – Data Interpretation & Structured Explanations: 20 × 4 Marks (80 Marks total)
- Section C – Core Definitions & Quantitative Calculations: 10 × 2 Marks (20 Marks total)
- Section D – 10 High-Frequency Core Repeated Questions for Exam Mastery & Knowledge Consolidation
- Division: 80% Authentic Cambridge 9700 Past Paper references + 20% Original A* Extension Problems

SYLLABUS SUBTOPICS COVERED IN THIS PACK

3.1 Mode of action of enzymes: globular proteins, intracellular vs extracellular enzymes, active site, enzyme-substrate (ES) and enzyme-product (EP) complexes, lowering of activation energy (E_a), lock-and-key vs induced-fit hypotheses, enzyme specificity, monitoring reaction progress (measuring product formation with catalase, substrate disappearance with amylase, colorimetric assays)

3.2 Factors that affect enzyme action: effects of temperature (kinetic energy, thermal denaturation), pH (buffers, charge alteration of R-groups), enzyme concentration (limiting factors), substrate concentration (saturation, V_{max}), Michaelis-Menten constant (K_m , substrate affinity), reversible inhibitors (competitive vs non-competitive, allosteric regulation), immobilised enzymes in alginate (thermal stability, reusability, continuous processing)

EXAMINER'S DIRECTIVE FOR CANDIDATE HAMNA

Write legibly in the dotted spaces provided. Pay strict attention to Cambridge command words (Explain, Describe, Deduce, Calculate).

Where biological diagrams or graphs are provided, extract exact qualitative and quantitative coordinates to support your answers.

1 9700/22/M/J/23/Q3 – Activation Energy Profile & Reaction Energetics

[6 Marks] Syllabus 3.1

Chemical reactions in biological systems require an initial input of energy to overcome the activation energy barrier. Fig. 3.1 shows the free energy profile of an exergonic reaction in the presence and absence of an enzyme.

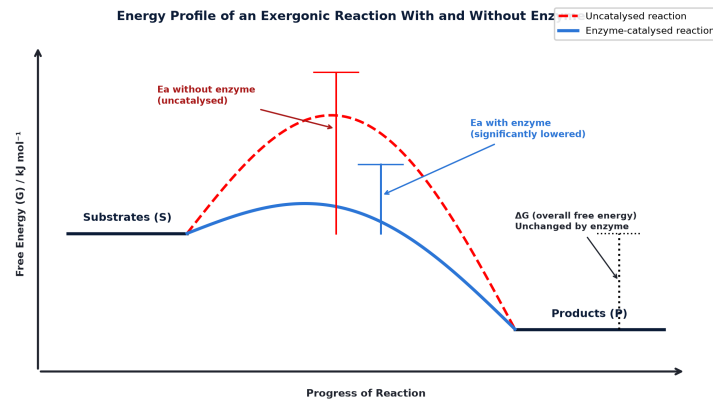


Fig. 3.1: Energy profile of an exergonic reaction with and without enzyme catalysis.

- (a) With reference to Fig. 3.1, define activation energy (E_a) and explain how the enzyme facilitates the conversion of substrates into products. [2]

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- (b) Explain why the overall free energy change (ΔG) of the reaction is identical for both the catalysed and uncatalysed pathways. [2]

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- (c) Describe how the formation of an enzyme-substrate (ES) complex lowers the activation energy of a metabolic reaction. [2]

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2 9700/21/O/N/22/Q3 – The Induced-Fit Mechanism of Enzyme Action

[6 Marks] Syllabus 3.1

Historically, Emil Fischer proposed the rigid 'lock-and-key' model, while Daniel Koshland refined this into the dynamic 'induced-fit' hypothesis. Fig. 3.2 compares these two models of catalytic mechanism.

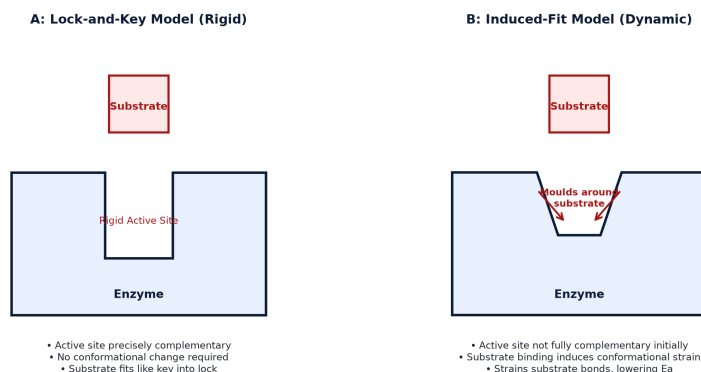


Fig. 3.2: Comparison of the rigid lock-and-key model (A) and dynamic induced-fit model (B).

- (a) Describe the fundamental difference between the lock-and-key model and the induced-fit model shown in Fig. 3.2. [2]

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- (b) Explain why the induced-fit model provides a better explanation for how enzymes lower the activation energy of a chemical reaction. [2]

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- (c) After product formation, explain what happens to the conformation of the active site. [2]

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3 9700/22/F/M/23/Q4 – Catalase Kinetics & Initial Rate Determination

[6 Marks] Syllabus 3.1

A student investigated the decomposition of hydrogen peroxide (H_2O_2) by intracellular catalase extracted from yeast. The volume of oxygen gas (O_2) evolved was measured every 10 seconds using a gas syringe. Fig. 3.3 shows the progress curve obtained.

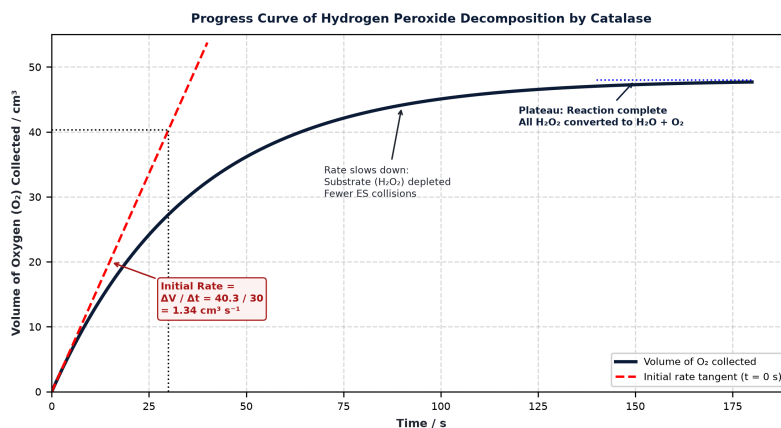


Fig. 3.3: Volume of oxygen gas evolved over time during catalase decomposition of hydrogen peroxide.

- (a) Write the balanced chemical equation for the reaction catalysed by catalase. [2]

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- (b) With reference to Fig. 3.3, determine the initial rate of reaction at $t = 0$ s, showing your working and stating appropriate units. [2]

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- (c) Explain why the rate of reaction gradually decreases as time progresses and eventually reaches a plateau. [2]

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4 9700/23/M/J/21/Q3 – Colorimetric Assay of Amylase Substrate Depletion

[6 Marks] Syllabus 3.1

The hydrolysis of starch by salivary amylase was monitored in a colorimeter. Samples were withdrawn at intervals, mixed with standardized iodine–potassium iodide solution, and percentage light transmission was measured at 580 nm. Fig. 3.4 shows the results.

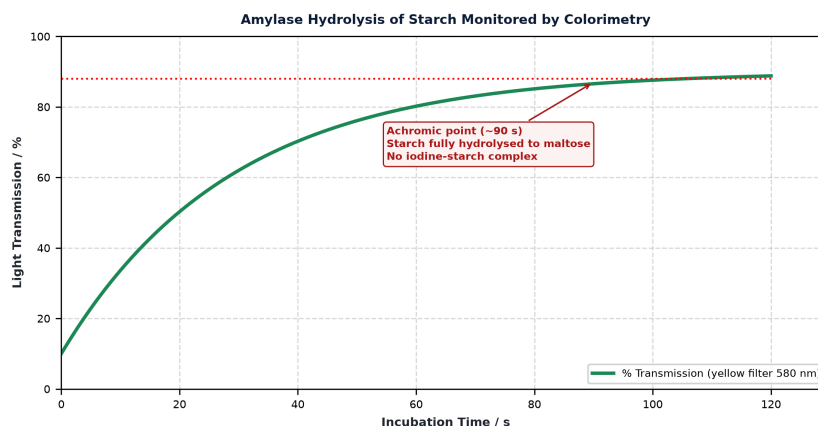


Fig. 3.4: Light transmission percentage vs incubation time during starch hydrolysis by amylase.

- (a) Explain why the percentage light transmission increases as the reaction proceeds. [2]
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- (b) With reference to Fig. 3.4, identify the achromic point of the reaction and explain its biochemical significance. [2]
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- (c) Suggest why using a colorimeter is superior to visual inspection of colour changes on a spotting tile. [2]
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5 9700/22/O/N/23/Q4 – Thermal Kinetics, Q10 & Irreversible Denaturation

[6 Marks] Syllabus 3.2

The activity of an enzyme was measured across a temperature range of 0 deg C to 75 deg C. Fig. 3.5 shows the resulting rate curve.

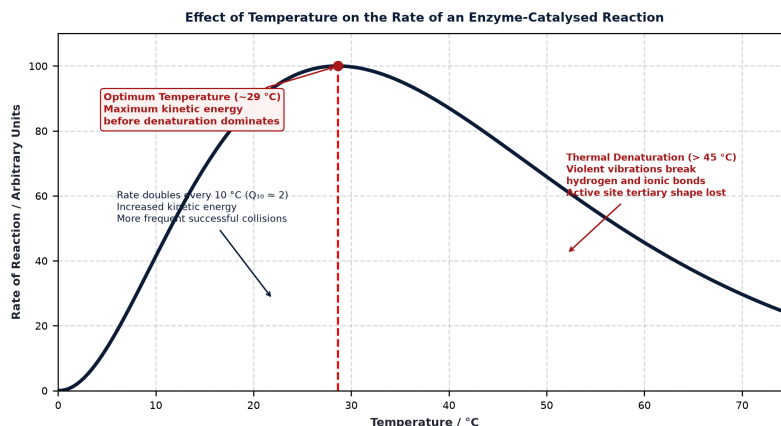


Fig. 3.5: Effect of temperature on initial reaction rate showing optimum and thermal denaturation.

- (a) Explain why the rate of reaction approximately doubles for every 10 deg C rise in temperature between 0 deg C and 30 deg C ($Q_{10} \approx 2$). [2]

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- (b) Identify the optimum temperature from Fig. 3.5 and define what is meant by optimum temperature. [2]

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- (c) Explain in molecular detail why the rate of reaction drops sharply above 45 deg C. [2]

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6 9700/21/M/J/22/Q3 – pH Sensitivity & Active Site Ionisation States

[6 Marks] Syllabus 3.2

Human digestive enzymes function in specialized anatomical compartments with distinct physiological pH levels. Fig. 3.6 compares the activity curves of pepsin, salivary amylase, and trypsin.

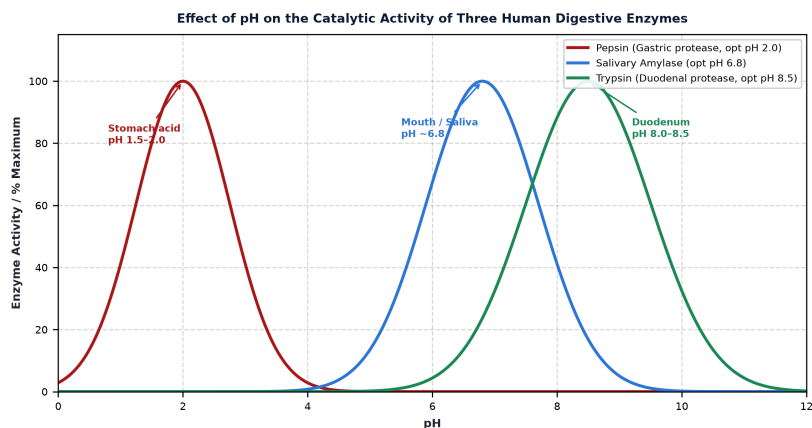


Fig. 3.6: Effect of pH on catalytic activity of pepsin, salivary amylase, and trypsin.

- (a) State the optimum pH for pepsin and trypsin from Fig. 3.6 and name the anatomical organ where each functions in vivo. [2]
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- (b) Explain how changes in pH away from the optimum alter the charge distribution of amino acid R-groups in the active site. [2]
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- (c) Explain why enzymes require buffer solutions when investigating the effect of substrate concentration on reaction rate. [2]
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7 9700/22/M/J/21/Q3 – Enzyme Concentration as a Rate-Limiting Factor

[6 Marks] Syllabus 3.2

The effect of enzyme concentration on initial rate was tested under two different substrate conditions: substrate in excess and substrate limited. Fig. 3.7 shows the two curves obtained.

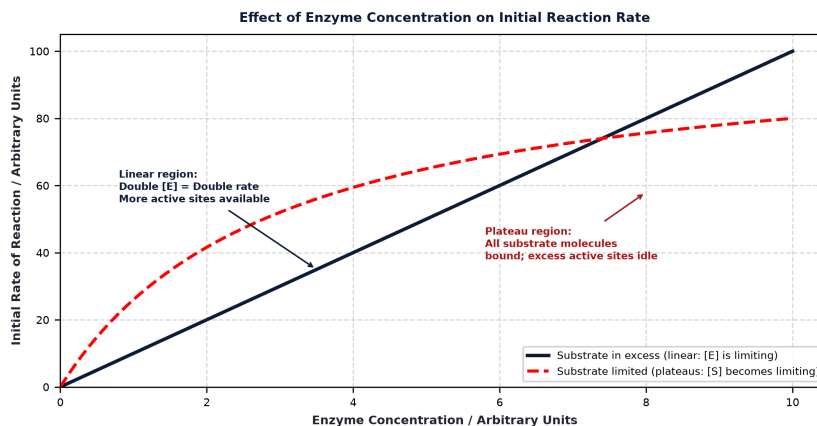


Fig. 3.7: Initial reaction rate vs enzyme concentration under excess and limiting substrate conditions.

- (a) Explain why the reaction rate is directly proportional to enzyme concentration when substrate is in excess. [2]

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- (b) Explain why the rate of reaction plateaus when substrate concentration is limited. [2]

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- (c) Suggest how you could experimentally test whether substrate concentration is limiting at the plateau. [2]

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8 9700/23/O/N/22/Q2 – Substrate Saturation & Michaelis-Menten Kinetics

[6 Marks] Syllabus 3.2

When the concentration of an enzyme is held constant and substrate concentration is increased, the reaction rate follows a characteristic hyperbolic curve. Fig. 3.8 shows this saturation curve.

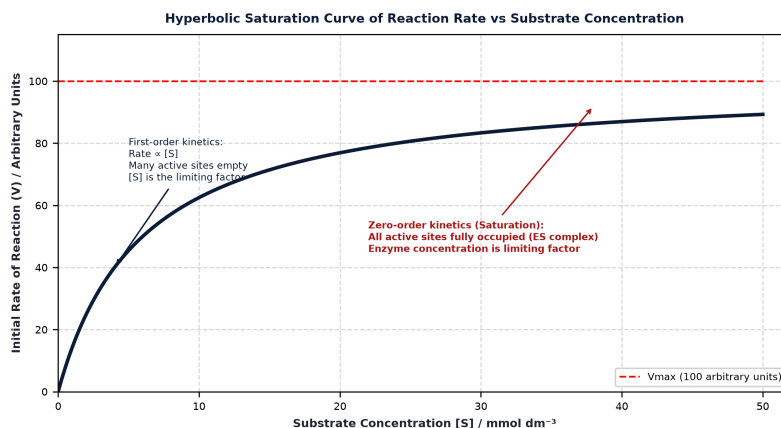


Fig. 3.8: Hyperbolic curve of initial reaction rate (V) vs substrate concentration [S].

- (a) Explain why the reaction rate increases steeply at low substrate concentrations (first-order kinetics). [2]

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- (b) Explain why increasing substrate concentration beyond 40 mmol dm⁻³ produces no further increase in reaction rate (zero-order kinetics). [2]

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- (c) State the single modification to the reaction mixture that would increase the maximum rate (V_{max}). [2]

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9 9700/22/F/M/22/Q4 – Derivation of K_m & Comparing Enzyme Affinities

[6 Marks] Syllabus 3.2

The Michaelis–Menten constant (K_m) is a fundamental kinetic parameter in enzymology. Fig. 3.9 shows the kinetic curves of two enzymes, Enzyme 1 and Enzyme 2, that act on the same substrate.

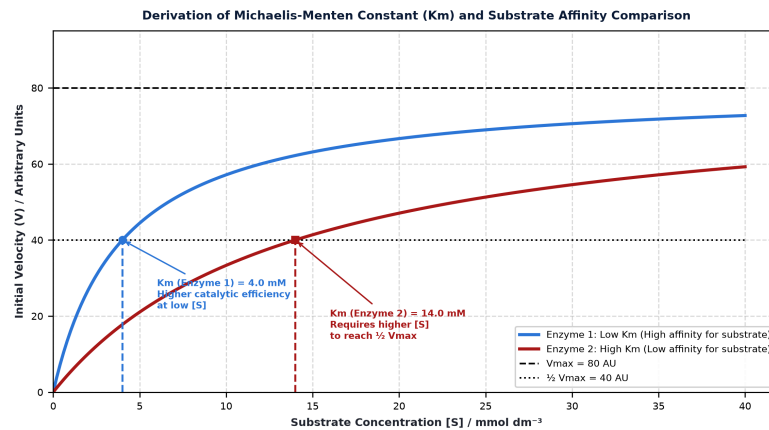


Fig. 3.9: Derivation of K_m from $\frac{1}{2} V_{max}$ for Enzyme 1 and Enzyme 2.

- (a) Define the Michaelis–Menten constant (K_m) in terms of V_{max} . [2]
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- (b) Using Fig. 3.9, determine the numerical value of K_m for Enzyme 1 and Enzyme 2. [2]
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- (c) State which enzyme has the higher affinity for the substrate and explain the physiological advantage of having a low K_m value in low substrate environments. [2]
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10 9700/21/M/J/23/Q2 – Competitive Enzyme Inhibition & Reversible Kinetics

[6 Marks] Syllabus 3.2

Malonate is a classic competitive inhibitor of the mitochondrial enzyme succinate dehydrogenase. Fig. 3.10 shows the mechanism and the effect of a competitive inhibitor on reaction rate.

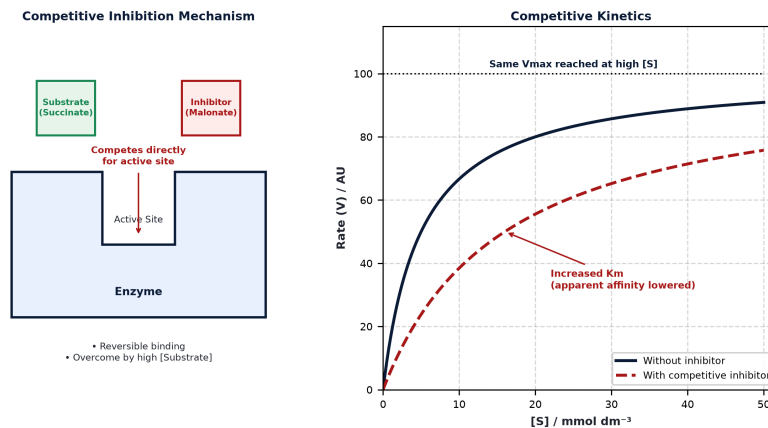


Fig. 3.10: Molecular mechanism and kinetic curve of competitive inhibition.

- (a) Describe how the molecular structure of a competitive inhibitor relates to that of the normal substrate. [2]

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- (b) With reference to Fig. 3.10, explain why a competitive inhibitor increases the apparent Km of an enzyme without altering Vmax. [2]

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- (c) Explain how the effect of a competitive inhibitor can be experimentally overcome. [2]

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11 9700/22/O/N/21/Q3 – Non-Competitive Inhibition & Allosteric Site Regulation

[6 Marks] Syllabus 3.2

Non-competitive inhibitors do not compete directly for the active site. Fig. 3.11 illustrates the allosteric mechanism and its effect on reaction kinetics.

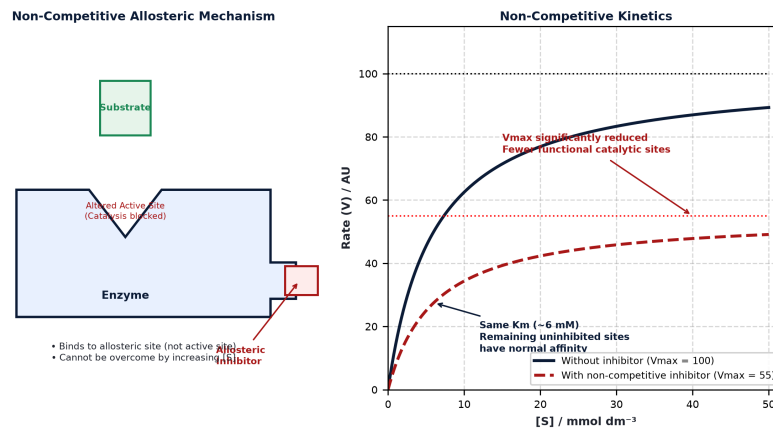


Fig. 3.11: Non-competitive inhibitor binding to allosteric site and resulting kinetics.

- (a) Describe where a non-competitive inhibitor binds on an enzyme molecule and how this alters active site function. [2]

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- (b) Explain why increasing substrate concentration cannot restore the original Vmax in the presence of a non-competitive inhibitor. [2]

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- (c) Explain why the value of Km remains unchanged in the presence of a pure non-competitive inhibitor. [2]

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12 9700/22/M/J/20/Q3 – Diagnostic Analysis via Lineweaver-Burk Plots

[6 Marks] Syllabus 3.2

The Lineweaver-Burk plot linearises Michaelis-Menten data by plotting the reciprocal of velocity ($1/v$) against the reciprocal of substrate concentration ($1/[S]$). Fig. 3.12 compares normal, competitive, and non-competitive enzyme kinetics.

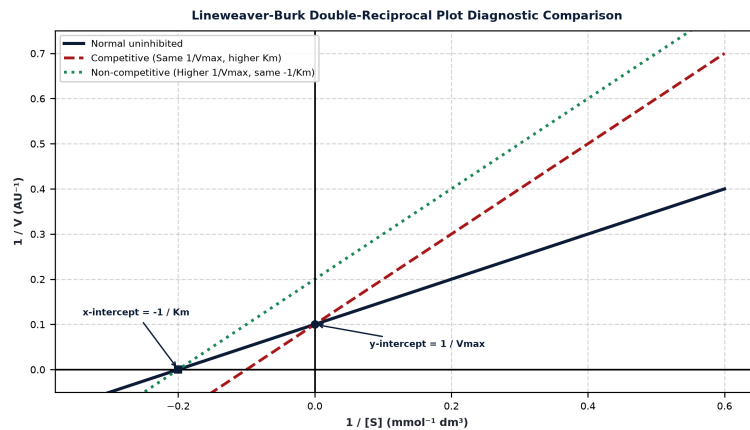


Fig. 3.12: Lineweaver-Burk double-reciprocal plot comparing uninhibited and inhibited enzymes.

- (a) State the kinetic meaning of the y-intercept and x-intercept on a Lineweaver-Burk plot. [2]

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- (b) With reference to Fig. 3.12, explain how a Lineweaver-Burk plot distinguishes between a competitive and a non-competitive inhibitor. [2]

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- (c) A line has a y-intercept of 0.20 AU⁻¹ and an x-intercept of -0.25 mmol⁻¹ dm³. Calculate the values of V_{max} and K_m for this enzyme. [2]

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13 9700/23/M/J/22/Q3 – Industrial Biotechnology: Immobilised Enzymes in Alginate

[6 Marks] Syllabus 3.2

In biotechnology, enzymes are frequently immobilized within porous calcium alginate beads. Fig. 3.13 shows a packed-bed bioreactor column and the thermal stability comparison between free and immobilised lactase.

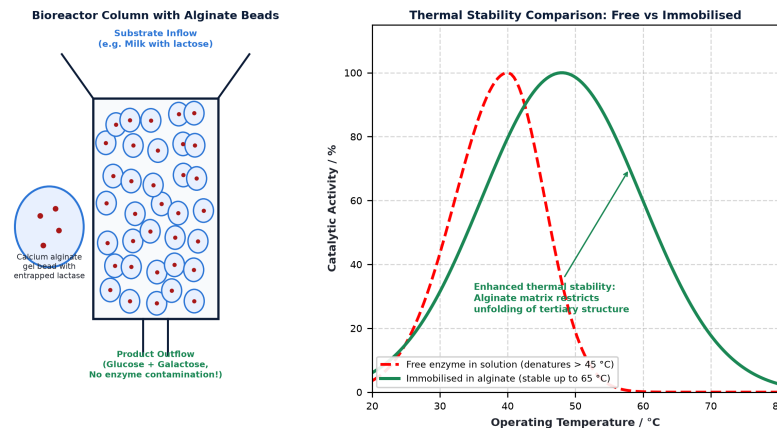


Fig. 3.13: Packed-bed bioreactor column (left) and thermal stability curves of free vs immobilised lactase (right).

- (a) Describe how lactase is immobilised into calcium alginate beads in the laboratory. [2]
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- (b) Explain two commercial advantages of using immobilised enzymes in a continuous bioreactor column compared to free enzymes. [2]
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- (c) With reference to Fig. 3.13, explain why the immobilised enzyme retains high catalytic activity at 60 deg C while the free enzyme is completely inactivated. [2]
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14 9700/21/O/N/23/Q4 – End-Product Allosteric Feedback Control

[6 Marks] Syllabus 3.2

Multi-step biochemical pathways are regulated to maintain cellular homeostasis without wasting chemical energy. Fig. 3.14 illustrates end-product inhibition in the synthesis of L-isoleucine from L-threonine.

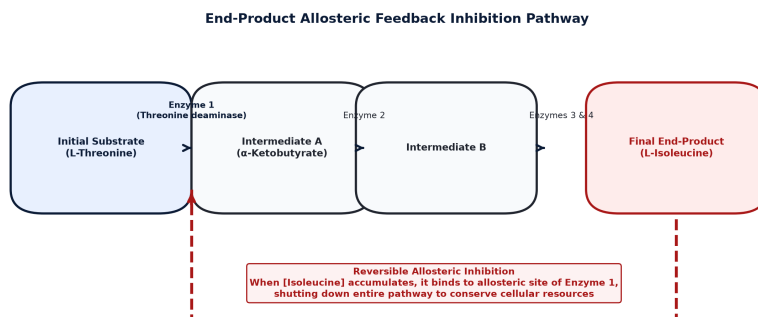


Fig. 3.14: Feedback inhibition of threonine deaminase by the end-product L-isoleucine.

- (a) With reference to Fig. 3.14, describe how accumulation of L-isoleucine down-regulates its own synthesis. [2]

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- (b) Explain why inhibiting the first enzyme in a biosynthetic pathway is more metabolically efficient than inhibiting an enzyme later in the pathway. [2]

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- (c) Explain what happens to the activity of threonine deaminase when intracellular levels of isoleucine decline. [2]

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15 9700/22/F/M/21/Q4 – Intracellular vs Extracellular Enzymes

[6 Marks] Syllabus 3.1

Enzymes operate both within cytoplasmic compartments and in the extracellular environment.

- (a) Distinguish between an intracellular enzyme and an extracellular enzyme, providing one named biological example of each. [2]

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- (b) Explain why intracellular lysosomal enzymes (acid hydrolases) have an optimum pH of ~5.0, whereas cytoplasmic enzymes have an optimum pH of ~7.2. [2]

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- (c) Describe the cellular pathway by which extracellular enzymes are synthesized and secreted from eukaryotic cells. [2]

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16 9700/21/M/J/20/Q3 – Experimental Protocol for K_m and V_{max} Determination

[6 Marks] Syllabus 3.2

A researcher designed an investigation to determine the kinetic parameters K_m and V_{max} of a novel bacterial cellulase.

- (a) Outline the experimental procedure used to measure the initial reaction velocity across a range of substrate concentrations. [2]
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- (b) Explain why the initial velocity (V_0) must be measured rather than the total product formed after 30 minutes. [2]
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- (c) State two environmental variables that must be strictly controlled throughout the investigation. [2]
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17 9700/23/O/N/21/Q2 – Structural Flexibility & Catalytic Transition State

[6 Marks] Syllabus 3.1

Advances in structural biology have provided direct evidence for conformational changes during catalysis.

- (a) Explain how X-ray crystallography of unliganded vs substrate-bound enzymes supports the induced-fit hypothesis. [2]
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- (b) Describe the role of catalytic residues within the active site compared to binding residues. [2]
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- (c) Explain why transition state analogues act as exceptionally potent enzyme inhibitors. [2]
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18 9700/22/M/J/22/Q4 – Industrial Bioprocessing: Free vs Immobilised Systems

[6 Marks] Syllabus 3.2

Industrial biocatalysis utilizes both soluble batch systems and immobilised continuous-flow columns.

- (a) Compare the economic cost of enzyme replacement in batch fermentation with free enzymes versus continuous-flow columns with immobilised enzymes. [2]
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- (b) Explain how immobilisation alters the downstream processing and purity of the pharmaceutical product. [2]
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- (c) Suggest one disadvantage or limitation of immobilising an enzyme within an alginate matrix. [2]
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19 [Mentora Original A* Extension] Q19 – Quantitative Analysis of the Michaelis–Menten Equation

[6 Marks] Syllabus 3.2

The rate of an enzyme-catalysed reaction is described mathematically by the Michaelis–Menten equation: $V = (V_{\max} * [S]) / (K_m + [S])$.

- (a) Calculate the initial reaction velocity (V) as a fraction or percentage of $V_{\max}$ when $[S] = 2 K_m$. [2]

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- (b) Calculate the initial reaction velocity (V) as a fraction or percentage of $V_{\max}$ when $[S] = 0.5 K_m$. [2]

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- (c) Show mathematically that when substrate concentration is extremely high ($[S] \gg K_m$), the reaction velocity approaches $V_{\max}$. [2]

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20 [Mentora Original A* Extension] Q20 – Transition State Stabilization & Catalytic Mechanisms

[6 Marks] Syllabus 3.1

Enzyme active sites deploy precise chemical mechanisms to stabilise the transition state of substrate molecules.

- (a) Explain how general acid–base catalysis in an enzyme active site facilitates bond cleavage. [2]

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- (b) Explain the role of metal ion cofactors (such as Zn^{2+} in carbonic anhydrase) in polarising substrate bonds. [2]

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- (c) Explain why enzymes lose catalytic activity if a single amino acid residue in the active site is mutated, even if overall folding is preserved. [2]

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21 9700/22/O/N/22/Q3 – Active Site Anatomy: Binding vs Catalytic Residues

[4 Marks] Syllabus 3.1

An enzyme active site is a specialized cleft formed by the precise 3D folding of the polypeptide chain.

- (a) Explain why amino acid residues that form the active site may be widely separated in the primary structure. [2]

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- (b) Distinguish between the function of binding residues and catalytic residues within the active site. [2]

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22 9700/21/O/N/21/Q3 – Maxwell-Boltzmann Molecular Energy Distribution

[4 Marks] Syllabus 3.1

At any given temperature, molecules in a solution possess a range of kinetic energies.

- (a) Describe how lowering the activation energy alters the proportion of substrate molecules capable of reacting. [2]

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- (b) Explain why an uncatalysed reaction occurs extremely slowly at 37 deg C even if it is energetically exergonic. [2]

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23 9700/23/M/J/23/Q4 – Reversible vs Irreversible Enzyme Inhibition

[4 Marks] Syllabus 3.2

Inhibitors are classified by the nature and permanence of their binding to enzymes.

- (a) Distinguish between reversible and irreversible enzyme inhibition in terms of bonding. [2]

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- (b) Explain why organophosphate nerve agents that covalently bind to acetylcholinesterase are lethal toxins. [2]

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24 9700/22/F/M/20/Q3 – Chemical Buffers & Experimental Standardisation

[4 Marks] Syllabus 3.2

Enzymatic investigations require rigorous control of physical and chemical variables.

- (a) Explain how a buffer solution maintains a relatively constant pH. [2]

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- (b) Explain why failing to use a buffer when testing the effect of substrate concentration invalidates the results. [2]

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25 9700/21/M/J/21/Q4 – Theoretical Rationale for Measuring Initial Rates

[4 Marks] Syllabus 3.1

In enzyme kinetics, measurements must always be taken at the start of the reaction.

- (a) Define what is meant by the initial rate of reaction (V_0). [2]

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- (b) Explain two reasons why the rate of reaction declines rapidly after the initial period. [2]

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26 9700/22/O/N/20/Q4 – Physiological Significance of the Michaelis-Menten Constant

[4 Marks] Syllabus 3.2

Different enzymes catalysing the same reaction can possess dramatically different K_m values.

- (a) Explain why K_m is inversely proportional to the affinity of an enzyme for its substrate. [2]

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- (b) Liver glucokinase has a high K_m (10 mM) for glucose, whereas brain hexokinase has a low K_m (0.1 mM). Relate this to the physiological function of each organ. [2]

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27 9700/23/M/J/20/Q3 – Allosteric Cooperativity & Sigmoidal Kinetics

[4 Marks] Syllabus 3.2

Certain regulatory enzymes exhibit cooperativity rather than standard hyperbolic Michaelis-Menten kinetics.

- (a) Describe the shape of the velocity versus substrate curve for an allosteric enzyme exhibiting positive cooperativity. [2]

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- (b) Explain how substrate binding to one subunit affects the remaining subunits in a multi-subunit allosteric enzyme. [2]

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28 9700/22/M/J/23/Q5 – Polymer Chemistry of Alginate Immobilisation

[4 Marks] Syllabus 3.2

Sodium alginate is a polysaccharide extracted from brown seaweed used to encapsulate enzymes.

- (a) Explain the chemical cross-linking that occurs when sodium alginate drops into calcium chloride solution. [2]

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- (b) Explain why high-molecular-weight substrates cannot be easily hydrolysed by enzymes entrapped in alginate beads. [2]

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29 9700/21/O/N/23/Q5 – Heavy Metal Toxicity & Disruption of Disulfide Bridges

[4 Marks] Syllabus 3.2

Heavy metal cations such as mercury (Hg^{2+}), lead (Pb^{2+}), and silver (Ag^+) act as non-competitive enzyme inhibitors.

- (a) Explain the chemical mechanism by which heavy metal ions disrupt protein tertiary structure. [2]

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- (b) Explain why poisoning by heavy metals cannot be reversed by administering excess substrate. [2]

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30 9700/22/F/M/22/Q5 – Industrial Production of Lactose-Reduced Dairy Products

[4 Marks] Syllabus 3.2

Lactase hydrolyses the milk disaccharide lactose into glucose and galactose.

- (a) State two nutritional and commercial benefits of treating milk with lactase. [2]

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- (b) Explain why lactose-free milk tastes noticeably sweeter than untreated regular milk. [2]

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31 9700/23/O/N/22/Q5 – Diagnostic Kinetic Behaviour at High Substrate Levels

[4 Marks] Syllabus 3.2

Enzymologists distinguish between inhibitor types by examining reaction rates at very high substrate concentrations.

- (a) Explain what happens to the rate of a competitive-inhibited reaction when substrate concentration is increased to saturation. [2]

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- (b) Explain what happens to the rate of a non-competitive-inhibited reaction under the same conditions. [2]

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32 9700/22/M/J/21/Q5 – Ecological Role of Extracellular Hydrolases

[4 Marks] Syllabus 3.1

Fungi and bacteria play a vital role in nutrient cycling as saprotrophic decomposers.

- (a) Describe how saprotrophic fungi utilize extracellular enzymes to obtain nutrients from decaying wood. [2]

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- (b) Explain why saprotrophs must secrete enzymes rather than ingesting whole macromolecules by endocytosis. [2]

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33 9700/21/M/J/23/Q5 – Substrate Specificity of Amylase on Branched Starch

[4 Marks] Syllabus 3.1

Salivary alpha-amylase is an endoglycosidase that hydrolyses internal bonds in starch.

- (a) State which specific glycosidic bonds are cleaved by alpha-amylase. [2]

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- (b) Explain why alpha-amylase cannot completely hydrolyse amylopectin into maltose, naming the intermediate products formed. [2]

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34 9700/22/O/N/22/Q5 – Colorimetric Calibration in Enzyme Assays

[4 Marks] Syllabus 3.1

Quantitative colorimetry requires an initial calibration curve using known standards.

- (a) Describe how a calibration curve of absorbance versus starch concentration is constructed. [2]

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- (b) Explain why colorimeter cuvettes must be wiped clean and inserted with the optical face correctly aligned. [2]

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35 9700/21/O/N/22/Q5 – Low Temperature Inactivation vs Denaturation

[4 Marks] Syllabus 3.2

Food can be preserved by freezing at -18°C .

- (a) Explain why enzymes are completely inactive at -18°C . [2]

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- (b) Explain why freezing an enzyme does not denature it, in contrast to boiling. [2]

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36 9700/23/M/J/22/Q5 – Pharmacology of Statins as Competitive Inhibitors

[4 Marks] Syllabus 3.2

Statins are widely prescribed medications used to reduce the risk of cardiovascular disease.

- (a) Describe the mechanism by which statins inhibit HMG-CoA reductase in hepatic cholesterol biosynthesis. [2]

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- (b) Explain how competitive inhibition by statins lowers low-density lipoprotein (LDL) cholesterol in the bloodstream. [2]

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37 [Mentora Original A* Extension] Q37 – The Quasi-Steady-State Assumption in Enzyme Kinetics

[4 Marks] Syllabus 3.2

The derivation of Michaelis-Menten kinetics depends on the Briggs-Haldane steady-state approximation.

- (a) State the core assumption of the quasi-steady-state hypothesis regarding the enzyme-substrate complex [ES]. [2]

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- (b) Explain what condition must be met regarding initial substrate concentration [S] relative to total enzyme concentration [E]_{total} for steady-state kinetics to hold. [2]

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38 [Mentora Original A* Extension] Q38 – Catalytic Constant (k_{cat}) & Catalytic Perfection

[4 Marks] Syllabus 3.2

The catalytic turnover number (k_{cat}) measures how rapidly an enzyme converts substrate into product.

- (a) Define the turnover number (k_{cat}) and explain how it is calculated from V_{max} and total enzyme concentration [E]_{total}. [2]

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- (b) The enzyme catalase has a k_{cat}/K_m ratio approaching 10⁸ dm³ mol⁻¹ s⁻¹. Explain what is meant by the statement that catalase has reached 'catalytic perfection'. [2]

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39 [Mentora Original A* Extension] Q39 – Uncompetitive Inhibition Mechanism

[4 Marks] Syllabus 3.2

In uncompetitive inhibition, the inhibitor cannot bind to free enzyme, but binds exclusively to the enzyme-substrate (ES) complex.

- (a) Explain how uncompetitive inhibitor binding prevents product formation. [2]

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- (b) Explain why both V_{max} and apparent K_m decrease by the same factor in uncompetitive inhibition. [2]

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40 [Mentora Original A* Extension] Q40 – Experimental Error Analysis: Thermal Equilibration

[4 Marks] Syllabus 3.2

In investigations of temperature effects on enzyme rates, systematic errors frequently arise from inadequate pre-incubation.

- (a) Explain why both the enzyme solution and substrate solution must be pre-incubated separately in a water bath before mixing. [2]

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- (b) Explain how mixing solutions that have not reached thermal equilibrium produces an incorrect rate measurement. [2]

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41 9700/22/F/M/23/Q2 – Definition of an Enzyme

[2 Marks] Syllabus 3.1

Enzymes underpin biological catalysis.

- (a) Define the term enzyme and state the macromolecular class to which nearly all enzymes belong. [2]

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42 9700/21/M/J/22/Q2 – Intracellular vs Extracellular Localization

[2 Marks] Syllabus 3.1

Enzyme location reflects functional role.

- (a) Distinguish between intracellular and extracellular enzymes, giving one named example of each. [2]

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43 9700/22/M/J/22/Q2 – Definition of an Active Site

[2 Marks] Syllabus 3.1

Enzymes have a specific active region.

- (a) Define the active site of an enzyme. [2]

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44 9700/23/M/J/21/Q2 – Definition of Activation Energy

[2 Marks] Syllabus 3.1

Chemical reactions require an activation barrier.

- (a) Define activation energy (E_a). [2]

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45 9700/22/O/N/21/Q2 – Competitive Inhibitor Effects on Kinetic Constants

[2 Marks] Syllabus 3.2

Inhibitors affect kinetic constants characteristically.

- (a) State the effect of a competitive inhibitor on V_{max} and on K_m . [2]

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46 9700/21/O/N/21/Q2 – Non-Competitive Inhibitor Effects on Kinetic Constants

[2 Marks] Syllabus 3.2

Allosteric inhibitors alter kinetics differently.

- (a) State the effect of a non-competitive inhibitor on V_{max} and on K_m . [2]

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47 9700/22/M/J/21/Q2 – Significance of a Low K_m Value

[2 Marks] Syllabus 3.2

K_m measures affinity.

- (a) Explain why a low value of K_m indicates a high affinity of an enzyme for its substrate. [2]

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48 9700/23/O/N/20/Q2 – Cross-Linking Reagent in Alginate Entrapment

[2 Marks] Syllabus 3.2

Enzyme immobilisation requires ionic cross-linking.

- (a) Name the salt solution into which sodium alginate is dropped to form insoluble gel beads. [2]

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49 9700/22/F/M/20/Q2 – Two Advantages of Immobilised Enzymes

[2 Marks] Syllabus 3.2

Immobilised enzymes offer processing advantages.

- (a) State two practical advantages of using immobilised enzymes in industry compared to free enzymes in solution. [2]

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50 [Mentora Original A* Extension] Q50 – Definition of an Allosteric Site

[2 Marks] Syllabus 3.2

Allosteric regulation controls metabolic flow.

- (a) Define the term allosteric site and explain its role in metabolic regulation. [2]

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SECTION D: 10 HIGH-FREQUENCY EXAMINER FAQs & CRITICAL REVISION PITFALLS

Essential Cambridge International AS Level Biology (9700) Mark-Scheme Traps, Core Misconceptions, and Model Answers

ENZYME MECHANISMS • ACTIVATION ENERGY (EA)

FAQ 1: Explain How Enzymes Lower Activation Energy and Contrast the Lock-and-Key with the Induced-Fit Hypothesis.

Cambridge Examiner Warning: Asserting that enzymes 'provide energy' to drive the reaction, or failing to state that the active site is flexible and changes conformation in the induced-fit model.

- Nature of Enzymes: Globular proteins acting as biological catalysts; possess a specific 3D active site composed of a few catalytic amino acid residues determined by tertiary structure.
- Activation Energy (E_a): The minimum kinetic energy required for colliding substrate molecules to reach the unstable transition state and react; enzymes lower E_a by providing an alternative reaction pathway.
- Mechanism of E_a Reduction: Binding in the active site brings substrates into close physical proximity and optimal stereochemical orientation; active site residues interact with substrate bonds, placing strain on covalent bonds to facilitate bond breaking.
- Lock-and-Key Hypothesis: The active site is considered rigid and perfectly complementary in shape to the substrate prior to binding; substrate fits precisely like a key in a lock.
- Induced-Fit Hypothesis: The active site is flexible; upon initial weak collision, the substrate induces a conformational change in the enzyme's tertiary structure, molding the active site snugly around the substrate, maximizing catalytic contact and bond strain.

REACTION KINETICS • GRAPHICAL ANALYSIS

FAQ 2: How and Why is the Initial Rate of Reaction Determined from Progress Curves (Tangent at $t = 0$)?

Cambridge Examiner Warning: Calculating average rate over several minutes rather than drawing a tangent at time zero, or failing to construct a sufficiently large right-angled triangle.

- Why Measure Initial Rate: At time $t = 0$, substrate concentration is known, at its maximum, and non-limiting; no product accumulation has occurred to cause product inhibition or reverse reactions; represents the true catalytic velocity.
- Curve Progression: Rate is steepest at $t = 0$; slows progressively as substrate is consumed (fewer effective substrate-active site collisions per unit time); eventually plateaus when all substrate is converted into product.
- Tangent Construction: Place a ruler along the initial linear section of the progress curve at the origin ($t = 0$); draw a straight line extending across both axes.
- Gradient Calculation: Construct a large right-angled triangle; calculate gradient = Change in Product (or Substrate) / Change in Time (dy/dx); include precise units (e.g. $\text{cm}^3 \text{s}^{-1}$ or $\text{mol dm}^{-3} \text{s}^{-1}$).

ENZYME KINETICS • SUBSTRATE SATURATION

FAQ 3: Explain Why Increasing Substrate Concentration Initially Increases Reaction Rate but Eventually Reaches a Plateau (V_{max}).

Cambridge Examiner Warning: Stating that at high substrate concentration the enzyme is 'denatured' or 'used up'. Enzymes are biological catalysts that remain unchanged and reusable.

- Initial Increase (Limiting Factor: Substrate): At low substrate concentrations, enzyme active sites are in excess; increasing substrate concentration increases the frequency of successful collisions between substrate and active sites; more enzyme-substrate (ES) complexes form per unit time, resulting in a proportional linear increase in reaction rate.
- Plateau Region (V_{max}): As substrate concentration rises further, active sites become increasingly occupied.
- Saturation Point: A concentration is reached where every enzyme active site is continuously occupied and working at maximum catalytic velocity; as soon as product is released, a new substrate binds immediately.
- Limiting Factor Shift: Enzyme concentration becomes the sole limiting factor; further increases in substrate concentration produce no additional increase in rate; reaction operates at V_{max} .

MICHAELIS-MENTEN KINETICS • K_m & SUBSTRATE AFFINITY

FAQ 4: Define the Michaelis-Menten Constant (K_m), Describe Its Graphical Derivation, and Explain Its Significance Regarding Affinity.

Cambridge Examiner Warning: Defining K_m as 'half of V_{max} ' ($V_{\text{max}}/2$ is a velocity; K_m is the substrate concentration). Stating that a higher K_m represents higher affinity (it is an inverse relationship).

- Definition of K_m : The substrate concentration $[S]$ at which the rate of the enzyme-catalysed reaction is half of the maximum velocity ($1/2 V_{\text{max}}$).
- Graphical Derivation: Identify the horizontal asymptote on a rate vs $[S]$ graph to determine V_{max} ; calculate $1/2 V_{\text{max}}$; locate this value on the y-axis, project horizontally to intersect the curve, then project vertically down to read K_m on the x-axis (with units of concentration, e.g. mmol dm^{-3}).
- Affinity Relationship (Inverse): K_m is inversely proportional to the enzyme's affinity for its substrate:
 1. Low K_m : Enzyme has high affinity for substrate; reaches $1/2 V_{\text{max}}$ at very low substrate concentration; forms ES complexes readily;
 2. High K_m : Enzyme has low affinity for substrate; requires high substrate concentration to reach $1/2 V_{\text{max}}$.
- Independence: K_m is a kinetic constant characteristic of an enzyme-substrate pair under specified pH and temperature; it is independent of enzyme concentration.

INHIBITION KINETICS • REVERSIBLE INHIBITORS

FAQ 5: Distinguish Between Competitive and Non-Competitive Inhibition with Reference to Active Sites, V_{max} , and K_m .

Cambridge Examiner Warning: Claiming that competitive inhibitors alter active site shape, or that non-competitive inhibitors can be overcome by adding more substrate.

- Competitive Inhibition:
 1. Molecular Structure: Inhibitor has a chemical shape very similar to the normal substrate;
 2. Binding Site: Competes directly with substrate to bind reversibly to the active site, blocking substrate entry;
 3. Reversibility & Effect of $[S]$: Can be completely overcome by increasing substrate concentration (higher probability of substrate colliding with active site than inhibitor);
 4. Kinetic Constants: V_{max} is UNCHANGED (eventually reached at high $[S]$); K_m is INCREASED (apparent affinity is reduced).
- Non-Competitive Inhibition:
 1. Molecular Structure: Inhibitor has no structural resemblance to the substrate;
 2. Binding Site: Binds reversibly to an allosteric site (a site away from the active site);
 3. Mechanism: Binding alters the enzyme's tertiary structure, causing conformational change in the active site so substrate no longer fits or catalytic groups are misaligned;
 4. Effect of $[S]$: Cannot be overcome by increasing substrate concentration;
 5. Kinetic Constants: V_{max} is REDUCED; K_m is UNCHANGED (unaffected enzyme molecules retain normal affinity).

ENVIRONMENTAL FACTORS • THERMAL INACTIVATION

FAQ 6: Explain the Molecular Basis of Temperature Effects on Enzyme Action: Kinetic Energy, Q_{10} , and Thermal Denaturation.

Cambridge Examiner Warning: Saying enzymes 'die' or that covalent peptide bonds break during thermal denaturation. Only non-covalent (and disulfide) bonds in tertiary conformation break.

- Temperature Below Optimum:
 1. Temperature increases kinetic energy of both enzyme and substrate molecules;
 2. Molecules move faster, increasing collision frequency per unit time;
 3. A greater proportion of colliding molecules possess kinetic energy exceeding the activation energy (E_a);
 4. Temperature coefficient (Q_{10}) is approximately 2.0 (rate doubles for every 10 deg C rise).
- Optimum Temperature: The temperature at which the rate of ES complex formation is maximal (~37–40 deg C in mammals, ~70 deg C+ in thermophiles).
- Temperature Above Optimum (Denaturation):
 1. Excessive thermal vibration imparts kinetic energy greater than the bond energy of weak intramolecular bonds;
 2. Hydrogen bonds and ionic bonds maintaining the tertiary structure break;
 3. Polypeptide chain uncoils; 3D conformation of the active site is permanently altered;
 4. Active site is no longer complementary to substrate; no ES complexes can form; rate drops precipitously to zero (irreversible denaturation).

ENVIRONMENTAL FACTORS • PH & CHARGE ALTERATION

FAQ 7: How Does Alteration of pH Affect the Ionisation of Active Site Residues and Enzyme Conformation?

Cambridge Examiner Warning: Claiming that pH changes alter molecular kinetic energy. pH affects the ionic charge and bonding of amino acid R-groups.

- Optimum pH: The hydrogen ion concentration ($[H^+]$) at which the active site has the optimal charge and shape for substrate binding and catalysis.
- Effect of Low pH (High $[H^+]$):
 1. Excess H^+ ions bind to negatively charged carboxylate groups ($-COO^- + H^+ \rightarrow -COOH$) of aspartic acid and glutamic acid R-groups, neutralising them;
 2. Breaks ionic bonds and salt bridges stabilising the enzyme's tertiary structure;
 3. Alters electrostatic attraction between charged active site residues and charged groups on the substrate.
- Effect of High pH (Low $[H^+]$ / High $[OH^-]$):
 1. OH^- ions remove protons from positively charged amino groups ($-NH_3^+ + OH^- \rightarrow -NH_2 + H_2O$) of lysine and arginine R-groups;
 2. Disrupts ionic bonds, altering tertiary structure;
 3. Extreme pH deviations cause irreversible denaturation of the active site.
- Buffer Action: Biological buffers resist changes in pH by donating or accepting H^+ ions, maintaining enzyme function.

BIOTECHNOLOGY • ENZYME IMMOBILISATION

FAQ 8: Explain the Principles, Kinetic Advantages, and Industrial Applications of Immobilised Enzymes in Alginate Beads.

Cambridge Examiner Warning: Stating that immobilised enzymes have a higher initial reaction rate than free enzymes. Substrate diffusion into beads slows initial kinetics.

- Entrapment Technique: Enzyme solution is mixed with sodium alginate, then dropped via syringe into calcium chloride ($CaCl_2$) solution; calcium ions replace sodium, cross-linking alginate chains to form insoluble calcium alginate gel beads entrapping the enzyme molecules.
- Industrial Advantages:
 1. Pure Product: Enzyme is physically retained in the matrix; product contains no enzyme contamination, eliminating expensive downstream separation and purification steps;
 2. Reusability: Beads are easily filtered and reused across multiple consecutive batches, dramatically lowering operating costs;
 3. Continuous Processing: Beads can be packed into vertical column reactors for continuous automated substrate flow;
 4. Enhanced Stability: The rigid calcium alginate matrix holds the tertiary structure of the enzyme in place, restricting molecular vibrations and significantly increasing tolerance to high temperatures and extreme pH before denaturation occurs.

METABOLIC CONTROL • ALLOSTERIC REGULATION

FAQ 9: Describe the Mechanism and Biological Significance of Allosteric End-Product Inhibition in Metabolic Pathways.

Cambridge Examiner Warning: Confusing end-product inhibition with competitive inhibition. It is non-competitive and reversible, acting at an allosteric site.

- Metabolic Pathways: Cellular biochemical reactions occur in multi-step enzyme cascades (A → B → C → D → Final Product), each catalyzed by a specific enzyme.
- Committed Step: The first unique enzyme in the pathway possesses both an active site (for substrate A) and an allosteric regulatory site.
- Allosteric Mechanism: When the final end-product accumulates in excess, it binds reversibly to the allosteric site of the first enzyme;
- Conformational Change: Allosteric binding transmits mechanical strain throughout the enzyme, altering the tertiary structure and 3D shape of the active site so it can no longer bind substrate A;
- Negative Feedback Significance: Halts the entire pathway, preventing wasteful overproduction of intermediate metabolites and conservation of cellular energy (ATP) and raw materials;
- Dynamic Resumption: As the end-product is consumed by the cell, its concentration falls, causing it to dissociate from the allosteric site; enzyme reverts to active conformation and pathway resumes.

ENZYME ASSAYS • INTRACELLULAR VS EXTRACELLULAR

FAQ 10: Distinguish Between Intracellular and Extracellular Enzymes, and Contrast Assay Methods for Measuring Product Formation vs Substrate Disappearance.

Cambridge Examiner Warning: Stating that catalase is an extracellular enzyme. Catalase is strictly intracellular (located inside peroxisomes). Confusing substrate loss assays with product formation assays.

- Intracellular Enzymes: Synthesised inside the cell and function within cellular compartments or cytosol; Examples: Catalase (located in peroxisomes, decomposes toxic hydrogen peroxide: $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$), DNA polymerase, ATP synthase, respiratory enzymes.
- Extracellular Enzymes: Synthesised inside cells on RER, packaged into secretory vesicles by Golgi body, secreted via exocytosis to function in external environments; Examples: Digestive enzymes like salivary/pancreatic amylase (hydrolyses starch to maltose in gut lumen), pepsin, trypsin.
- Monitoring Product Formation (Catalase Assay): Hydrogen peroxide mixed with catalase in a closed boiling tube connected to a gas syringe or inverted measuring cylinder filled with water; volume of O_2 gas produced recorded at regular time intervals (e.g. every 15 s) to plot a product-time progress curve.
- Monitoring Substrate Disappearance (Amylase Assay): Amylase mixed with starch solution; samples removed at timed intervals (every 30 s) and added to spotting tile wells containing iodine solution; time taken to reach achromatic point (yellow-brown, no blue-black) recorded, or disappearance of starch tracked continuously in a colorimeter using transmission.

EXAMINER MARK SCHEME & MARKING GUIDELINES

TOPIC 3 – ENZYMES – Cambridge International AS Level Biology (9700)

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
1		Activation energy is the minimum kinetic energy required by colliding reactant molecules to reach the unstable transition state; the enzyme lowers the activation energy barrier so a greater proportion of substrate molecules possess sufficient energy to react at body temperature [2].	[2]
1		Delta G is the difference in free energy between initial substrate ground state and final product ground state; enzymes accelerate reaction rate by altering transition state pathway but do not alter the thermodynamic stability of substrates or products [2].	[2]
1		Substrate binding into the active site places mechanical strain on specific covalent bonds / brings reactive groups into precise mutual orientation / creates favourable microenvironment with catalytic R-groups [2].	[2]
2		In lock-and-key, active site is rigidly complementary to substrate before binding; in induced-fit, active site is flexible and undergoes a conformational change to wrap precisely around substrate upon initial collision [2].	[2]
2		Conformational change exerts physical torsional stress / strain on substrate bonds, distorting them towards transition state geometry and reducing energy required to break bonds [2].	[2]
2		Products have lower affinity for active site and dissociate; active site reverts back to its original relaxed resting conformation, ready to accept another substrate molecule [2].	[2]
3		$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ [2].	[2]
3		Gradient of tangent at $t = 0$ s: $\Delta V / \Delta t = 40.3 / 30$; Initial rate = $1.34\text{ cm}^3\text{ s}^{-1}$ (accept 1.30 to $1.38\text{ cm}^3\text{ s}^{-1}$ with units) [2].	[2]
3		Substrate (H_2O_2) is consumed, lowering substrate concentration; fewer successful collisions between substrate and active sites per unit time; plateau occurs when all H_2O_2 is completely converted into products [2].	[2]
4		Amylase hydrolyses starch into maltose; less starch available to form blue-black polyiodide inclusion complex, so less light is absorbed and more light is transmitted through solution [2].	[2]
4		Achromic point is reached at ~90 s (transmission reaches maximum plateau ~88–90%); represents complete hydrolysis of starch (no helical amylose chains remain to bind iodine) [2].	[2]
4		Provides quantitative, objective numerical data; eliminates subjective human error in judging subtle colour transitions [2].	[2]
5		Increased thermal energy increases kinetic energy of enzyme and substrate molecules; faster movement increases frequency of collisions and proportion of collisions possessing energy $\geq E_a$ [2].	[2]
5		Optimum temperature is ~40 deg C (accept 38–42 deg C); the temperature at which the rate of catalytic conversion is highest [2].	[2]
5		Violent thermal vibrations break weak hydrogen and ionic bonds stabilizing tertiary structure; active site loses specific complementary shape (denatures), substrate cannot bind, ES complexes cannot form [2].	[2]
6		Pepsin: optimum pH 2.0 (stomach / gastric juice); Trypsin: optimum pH 8.5 (duodenum / small intestine) [2].	[2]
6		Excess H^+ (low pH) protonates $-\text{COO}^-$ groups to $-\text{COOH}$; excess OH^- (high pH) deprotonates $-\text{NH}_3^+$ groups to $-\text{NH}_2$; breaks ionic bonds and destroys precise electrical charge complementarity required to bind substrate [2].	[2]
6		Substrates or reaction products can release or consume H^+ ions, changing pH; buffers resist pH fluctuations to ensure pH remains constant as a controlled variable [2].	[2]
7		More enzyme molecules provide more available active sites per unit volume; increases frequency of successful collisions and rate of ES complex formation [2].	[2]
7		Substrate molecules become the limiting factor; all substrate molecules are already bound into active sites, leaving surplus enzyme active sites unoccupied / idle [2].	[2]
7		Add extra substrate to the reaction mixture; if rate increases, substrate was limiting [2].	[2]
8		At low [S], most active sites are empty / unoccupied; increasing [S] increases collision frequency with active sites, so substrate concentration is rate-limiting [2].	[2]
8		All active sites are fully saturated with substrate; as soon as product leaves, another substrate immediately binds; maximum turnover rate is reached and enzyme concentration is limiting [2].	[2]
8		Increase the concentration of enzyme [2].	[2]
9		K_m is the substrate concentration at which the reaction velocity is half of its maximum rate ($1/2 V_{\text{max}}$) [2].	[2]
9		$V_{\text{max}} = 80\text{ AU} \rightarrow 1/2 V_{\text{max}} = 40\text{ AU}$; $K_m(\text{Enzyme 1}) = 4.0\text{ mmol dm}^{-3}$; $K_m(\text{Enzyme 2}) = 14.0\text{ mmol dm}^{-3}$ [2].	[2]
9		Enzyme 1 has higher affinity; can operate close to maximum catalytic capacity even when intracellular substrate concentrations are very low [2].	[2]
10		Competitive inhibitor has a molecular shape / geometry highly similar to the substrate, allowing it to fit into and bind reversibly to the active site [2].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
10		Inhibitor competes with substrate for active sites, requiring a higher substrate concentration to achieve $1/2 V_{max}$ (higher K_m); but at very high $[S]$, substrate outcompetes inhibitor so all active sites are saturated to reach normal V_{max} [2].	[2]
10		Significantly increase the concentration of the substrate; substrate molecules outnumber inhibitor molecules, vastly increasing probability of substrate binding [2].	[2]
11		Binds to an allosteric site (a site separate from the active site); alters tertiary conformation of the enzyme, distorting active site shape so catalysis cannot occur [2].	[2]
11		Inhibitor does not compete with substrate for active sites; inhibited enzyme molecules are functionally inactive regardless of how high substrate concentration is raised [2].	[2]
11		Enzyme molecules that remain uninhibited have unchanged, normal active sites and retain identical affinity for substrate, reaching half of their reduced V_{max} at the same $[S]$ [2].	[2]
12		y-intercept = $1 / V_{max}$; x-intercept = $-1 / K_m$ [2].	[2]
12		Competitive inhibitor shares same y-intercept (same V_{max}) but shifts x-intercept closer to origin (higher K_m); non-competitive inhibitor has higher y-intercept (lower V_{max}) but identical x-intercept (same K_m) [2].	[2]
12		$V_{max} = 1 / 0.20 = 5.0 \text{ AU}$; $K_m = -1 / (-0.25) = 4.0 \text{ mmol dm}^{-3}$ [2].	[2]
13		Mix lactase enzyme solution with sodium alginate; drop mixture using a syringe / pipette into a solution of calcium chloride (CaCl_2) to form insoluble spherical gel beads [2].	[2]
13		Enzyme is easily recovered and reused multiple times (lowers running costs); product is not contaminated with enzyme, eliminating downstream purification steps [2].	[2]
13		Alginate matrix physically holds enzyme in place and restricts molecular vibration / unfolding; stabilizing tertiary structure and active site conformation against thermal denaturation at higher temperatures [2].	[2]
14		Isoleucine binds reversibly to the allosteric site of Enzyme 1 (threonine deaminase); causes conformational change in active site that prevents threonine from binding, stopping production of intermediate A [2].	[2]
14		Prevents wasteful accumulation of toxic or redundant pathway intermediates (A and B); conserves ATP and substrate threonine for other vital metabolic pathways [2].	[2]
14		Isoleucine dissociates from the allosteric site; active site reverts to its active conformation and synthesis of isoleucine resumes (negative feedback homeostasis) [2].	[2]
15		Intracellular enzymes function inside cells (e.g. catalase, DNA polymerase); extracellular enzymes are secreted and function outside cells (e.g. pepsin, amylase) [2].	[2]
15		Inside lysosome, proton pumps maintain pH 5.0 for optimal hydrolysis of engulfed pathogens/organelles; if lysosome ruptures, enzymes are inactive in neutral cytoplasm (pH 7.2), protecting cell from autolysis [2].	[2]
15		Synthesized by ribosomes on rough endoplasmic reticulum → transported via vesicles to Golgi apparatus for modification/packaging → transport vesicles fuse with cell surface membrane (exocytosis) [2].	[2]
16		Prepare serial dilution of substrate concentrations; mix with constant enzyme concentration at controlled temperature/pH; measure product formed or substrate lost in first 30-60 seconds [2].	[2]
16		Initial rate reflects true kinetics when substrate concentration is known and un-depleted; over 30 minutes rate slows due to substrate depletion, product inhibition, or enzyme instability [2].	[2]
16		Temperature (using thermostatically controlled water bath); pH (using buffer solution) [2].	[2]
17		Shows atomic coordinates of active site loops and domain movements; active site cleft closes and catalytic residues shift closer to substrate upon binding [2].	[2]
17		Binding residues hold substrate in place via non-covalent interactions (hydrogen/ionic bonds); catalytic residues participate directly in bond-breaking/bond-making mechanisms [2].	[2]
17		Enzymes have evolved highest complementary affinity for transition state geometry rather than substrate; transition state analogues bind with extremely high affinity and block active site [2].	[2]
18		Batch free enzyme is discarded with product at end of cycle (high recurring enzyme cost); immobilised enzyme remains trapped in column and is reused for thousands of cycles (lowers cost) [2].	[2]
18		Product effluent is free of enzyme protein; avoids expensive chemical separation, filtration, and protein denaturation steps to purify product [2].	[2]
18		Substrate diffusion into bead interior is slower (diffusion limitation); alginate pores may leak enzyme over time or beads may degrade under shear stress [2].	[2]
19		Substitute $[S] = 2K_m$: $V = (V_{max} * 2K_m) / (K_m + 2K_m) = (2/3) * V_{max} = 0.67 V_{max}$ (66.7% of V_{max}) [2].	[2]
19		Substitute $[S] = 0.5K_m$: $V = (V_{max} * 0.5K_m) / (K_m + 0.5K_m) = (0.5/1.5) * V_{max} = (1/3) * V_{max} = 0.33 V_{max}$ (33.3% of V_{max}) [2].	[2]
19		When $[S] \gg K_m$, $(K_m + [S]) \approx [S]$; equation simplifies to $V \approx (V_{max} * [S]) / [S] = V_{max}$ (independent of $[S]$, zero-order kinetics) [2].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
20		Active site amino acid R-groups act as proton donors (e.g. protonated histidine) or proton acceptors; stabilizing charged intermediates during transition state [2].	[2]
20		Positively charged divalent metal ion coordinates with substrate water/carbonyl oxygen, drawing electron density away to increase nucleophilicity / polarise bond for rapid attack [2].	[2]
20		Loss of critical catalytic R-group eliminates specific chemical mechanism (e.g. nucleophilic attack or proton transfer); substrate may still bind but cannot undergo catalytic chemical transformation [2].	[2]
21		Polypeptide chain undergoes complex secondary coiling and tertiary folding; bringing distant linear residues into close spatial proximity in 3D space [2].	[2]
21		Binding residues recognize and anchor the substrate through complementary non-covalent bonds; catalytic residues act directly to break or form covalent bonds [2].	[2]
22		Shifts the activation energy threshold to the left on the energy distribution; a vastly greater fraction of colliding molecules have kinetic energy $\geq E_a$ [2].	[2]
22		The uncatalysed E_a is so high that only an infinitesimal fraction of molecules possess sufficient thermal kinetic energy at 37 deg C to surmount the activation barrier [2].	[2]
23		Reversible inhibitors bind via weak non-covalent interactions (H-bonds/ionic) and can dissociate; irreversible inhibitors form permanent covalent bonds with active site residues [2].	[2]
23		Permanently inactivate acetylcholinesterase; acetylcholine cannot be broken down at synapses, causing continuous uncontrolled muscle contraction and respiratory paralysis [2].	[2]
24		Contains a weak acid and its conjugate base; absorbs excess H^+ ions when acid is added and releases H^+ ions when base is added [2].	[2]
24		Any change in pH would alter the ionisation of active site R-groups; introducing an uncontrolled confounding variable that masks the true effect of [S] [2].	[2]
25		The rate of catalytic conversion at time zero ($t = 0$), measured from the tangent to the reaction progress curve at the immediate start [2].	[2]
25		Substrate concentration decreases as it is consumed, reducing collision frequency; accumulation of products may cause end-product inhibition [2].	[2]
26		A low K_m means half-maximal velocity is achieved at very low substrate concentration, showing strong substrate binding; high K_m requires high [S] to bind [2].	[2]
26		Hexokinase works at maximum rate even during hypoglycemia to ensure brain survival; glucokinase only phosphorylates glucose after carbohydrate-rich meals when blood glucose is elevated [2].	[2]
27		Sigmoidal (S-shaped) curve; initial low rate followed by a steep increase in rate over a narrow range of substrate concentrations [2].	[2]
27		Binding of substrate to one active site induces a conformational shift that transmits to adjacent subunits; increasing their affinity for substrate [2].	[2]
28		Divalent calcium ions (Ca^{2+}) displace monovalent sodium ions (Na^+); cross-linking carboxylate groups of adjacent alginate polymer chains into an insoluble 3D hydrogel mesh [2].	[2]
28		Large macromolecules (such as starch or insoluble proteins) cannot diffuse through the small pores of the alginate gel matrix to reach the trapped enzyme [2].	[2]
29		Heavy metal ions react strongly with sulfhydryl ($-SH$) groups of cysteine residues; disrupting covalent disulfide bridges and altering active site tertiary conformation [2].	[2]
29		Heavy metals bind to non-catalytic sites / distort active site directly; substrate cannot compete with heavy metals, so increasing [S] cannot restore activity [2].	[2]
30		Allows lactose-intolerant individuals to consume dairy without digestive distress; reduces risk of crystallization in condensed milk and ice cream [2].	[2]
30		Hydrolysis yields equal amounts of glucose and galactose; both monosaccharides have a higher relative sweetness rating on taste receptors than disaccharide lactose [2].	[2]
31		Rate reaches the normal uninhibited V_{max} ; high substrate concentration completely outcompetes inhibitor for active sites [2].	[2]
31		Rate plateaus at a lower maximum velocity; non-competitive inhibitor remains bound to allosteric site and inactivates a fraction of enzyme molecules [2].	[2]
32		Hyphae secrete cellulase and pectinase enzymes into surrounding organic matter; hydrolysing insoluble polymers into soluble glucose monomers that are absorbed by diffusion/active transport [2].	[2]
32		Fungal hyphae have rigid chitin cell walls that prevent phagocytosis / endocytosis; only small soluble molecules can traverse the cell wall and membrane [2].	[2]
33		Internal $\alpha(1 \rightarrow 4)$ -glycosidic bonds [2].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
33		Active site cannot bind or cleave alpha(1→6)-glycosidic bonds at branch points; leaves branched limit dextrins that require debranching enzyme for complete hydrolysis [2].	[2]
34		Prepare known standard starch concentrations; add constant volume of iodine reagent; measure absorbance at 580 nm and plot absorbance (y-axis) vs concentration (x-axis) [2].	[2]
34		Fingerprints or scratches on optical face scatter light; incorrect alignment alters path length, leading to erroneous absorbance measurements [2].	[2]
35		Water is frozen solid and molecules have extremely low kinetic energy; enzyme and substrate cannot diffuse or collide [2].	[2]
35		Freezing does not supply energy to break covalent, ionic, or hydrogen bonds; tertiary structure remains intact and enzyme resumes activity upon thawing [2].	[2]
36		Statins have a molecular structure similar to HMG-CoA substrate; bind competitively and reversibly to active site of HMG-CoA reductase, blocking mevalonate synthesis [2].	[2]
36		Decreased intracellular cholesterol synthesis prompts hepatocytes to upregulate LDL receptors; increasing clearance of LDL particles from systemic circulation [2].	[2]
37		The concentration of the enzyme-substrate complex [ES] remains constant over the measurement period (rate of ES formation = rate of ES breakdown) [2].	[2]
37		Initial substrate concentration must be vastly in excess of enzyme concentration ([S] >> [E]total); ensuring [ES] formation does not significantly deplete free substrate [2].	[2]
38		k _{cat} is the maximum number of substrate molecules converted to product per active site per second; calculated as k _{cat} = V _{max} / [E]total [2].	[2]
38		The reaction rate is limited solely by the physical rate of diffusion of substrate molecules colliding with the active site in aqueous solution (diffusion-controlled limit) [2].	[2]
39		Binds to a site created only after substrate binding; locks ES complex in an inactive non-productive ESI complex that cannot undergo catalysis to form product [2].	[2]
39		V _{max} decreases because active complexes are removed; apparent K _m decreases because depletion of ES pulls free E into ES (Le Chatelier's principle), apparently increasing affinity [2].	[2]
40		Ensures both reactants have reached the exact target test temperature prior to starting the reaction, so initial rate reflects specified temperature [2].	[2]
40		If mixed cold, initial rate will reflect a lower temperature during the crucial first 30 seconds before thermal equilibrium is attained, yielding an underestimated initial velocity [2].	[2]
41		A biological catalyst that accelerates the rate of metabolic reactions without being consumed or chemically altered; globular protein [2].	[2]
42		Intracellular enzymes act inside cells (e.g. catalase / DNA polymerase); extracellular enzymes are secreted and act outside cells (e.g. pepsin / amylase) [2].	[2]
43		The specific 3D region of an enzyme molecule, formed by tertiary folding, to which the substrate binds and where catalytic conversion occurs [2].	[2]
44		The minimum amount of energy required by colliding reactant molecules to overcome the energy barrier and initiate a chemical reaction [2].	[2]
45		V _{max} remains unchanged; K _m increases [2].	[2]
46		V _{max} decreases; K _m remains unchanged [2].	[2]
47		A lower substrate concentration is required to reach half the maximum rate (1/2 V _{max}), meaning the enzyme binds tightly to the substrate [2].	[2]
48		Calcium chloride solution (CaCl ₂) [2].	[2]
49		1. Enzyme can be easily recovered and reused; 2. Product is uncontaminated by enzyme / enzyme exhibits higher thermal stability [2].	[2]
50		A regulatory binding site distinct from the active site; binding of an effector molecule induces a conformational change that alters active site catalytic activity [2].	[2]